BUSA3021 Business Analytics Project, Session 2, 2026

Assessment Task	**Professional practice: Data analytics for business strategy**
Due date	Group report: Week 13 (Wed 4 Nov @ 11.55pm) Group presentations: in Week13 classes - Presented to the client (if selected): in Week 13 lecture (Tue 3 Nov, 1-3pm) OR - Presented to the teaching team in your allocated Week 13 tutorials **Important:** There is no option to record group presentations and/or present it individually or in group at a later date.
Weight (%)	40%: 30% Group report (one per group) and 10% Presentation (individual presentation marks)
Task description	Group report (one per group) and group presentation
Submission Method	Group report submitted via iLearn; Group presentation delivered **in-person** in Week 13: to the client in the lecture (if selected) or to the teaching team in the allocated tutorials
Feedback mechanism(s)	Rubric for both group report and group presentation; The client will provide oral feedback in Week 13 during in-class presentations; They may provide additional written feedback, which will be communicated to the groups via iLearn.
Feedback available (anticipated date)	This unit does not have the final exam so the feedback will be provided on iLearn on the MQ S2 marks release date.
Links to Unit Learning Outcomes	UL01, ULO2, ULO3, ULO4, ULO5

INTEGRITY MATTERS



Integrity matters – at home, in your workplace and here at University. As a highly valued member of the Macquarie University Community, you carry great responsibility to uphold the good name of our institution. Our reputation is your reputation and will stay with you for life when your degree opens amazing opportunities for you.

If you are ever unsure whether your actions fall within the guidelines of Academic Integrity, please don't hesitate to reach out. Contact the Academic Literacies Unit or your Tutor/Unit Convenor.

ASSESSMENT DESCRIPTION

Professional purpose: The 'Why' of this assignment

This assignment gives you an opportunity to work on a real-life project for the client and present your work in a written and oral form to different audiences. The hands-on project will help you gain a valuable (and highly marketable) experience of designing and implementing a Data Visualisation (DV) project, which combines design thinking, data management (including data modelling and data quality), data analysis, data ethics, visual ethics and state-of-the-art practice of considerations equity and diversity in DV. Your work will cover **a complete project lifecycle**, starting from data collection, implementation of a visual data exploration environment, all the way to the project delivery (including presentation) to the client. As you already know, communication and presentation skills are critical in any work setting. This project will help you to further improve these skills, even if you are already an experienced presenter. Finally, you will learn how to apply pattern thinking so you can recognise future opportunities to implement similar projects in *any* contexts, including organisational, cross-organisational, industry-wide or societal contexts.

Skills in focus for this assessment

Communication, Teamwork, Work-readiness, Design and Implementation of Visual Data Exploration Environment, Working with the client.

ASSESSMENT INSTRUCTIONS

Practical Project:

In this project you will design and implement an interactive visual data exploration environment for your client: Business Council of Cooperatives (BCCM) in their dual role: as the peak body for the co-operative & mutual (CME) enterprises in Australia and the Australian representative in the International Cooperative Alliance (ICA). Your interactive visual environment will reflect the client's dual role, by designing the corresponding 'personas', as per project description.

Project resources

To help you develop BA skills across **the whole lifecycle of a visual data exploration project**, you are not given a ready-made data set.

Instead, you are required to use ethical data collection (harvesting) method and construct high-quality data set from the *given* and *found* data sources, including both quantitative and qualitative data. Make sure that any additional data sources are reputable and relevant for this project.

Main Data Sources:

ICA World Cooperative Monitor **https://monitor.coop/**
(Note that the ICA Cooperative Monitor is different from the Cooperative World Map also hosted by ICA).

BCCM National Mutual Economy (NME) report https://bccm.coop/about/what-we-do/research/national-mutual-economy-report/

For ideas and inspiration, you may like to read about the other international initiatives related to cooperative maps or monitors around the world. Links are provided in the following ICA article: Counting Cooperatives Why Data Matters (ICA, Jan 2026)

Important considerations about data sources:

- Even though the given sources of data are public, you still need to adhere to the ethical use of public data, which includes non-commercial use of data and an acknowledgement of data sources. This means that your written report needs to include the reference to the main and other data sources.
- The above data set links take you to the latest ICA Cooperative Monitor & 2026 BCCM NME report. You are allowed and encouraged to access the Monitor and NMEs for the previous years in order to harvest data sets across different years.
- You are allowed to find and harvest additional data sources as long as they are **relevant to the project, reputable, public, not related to individual people/employees, and harvested using ethical methods.**
- Please note that BCCM is a different organisation from the Co-op Federation (previous BUSA3021 client). These two organisations have different mission/purpose and different member organisations. Consequently, it is not appropriate to harvest and use data from the Co-op Federation.

Design thinking:

The following design thinking references provide the necessary foundations for the design-thinking part of your project:

Personas in design thinking:

- https://en.wikipedia.org/wiki/Persona_(user_experience)
- http://www.servicedesigntools.org/tools/40
- http://www.slideshare.net/Frankichamaki/design-thinking-with-persona

Project specification

To design and implement a combined multi-organisational visual data exploration environment (VA application) for the client, you need to focus on decision-making needs of three different design personas, as specified below (in Task 1). Design personas are different types of users grouped together into the so-called "archetypes" (Godwin, 2009). Archetypes are typically built/designed after in-depth observations of work practices and interviews with potential users. Each resulting persona is designed to represent a fictional character whose profile sums up the features of a whole (existing) group of similar users, for example in a similar professional role (Godwin, 2009). Therefore, each persona

represents a much more than a typical user. Consequently, by practicing designing thinking in this project, you will go well beyond a typical software design phase called 'elicitation of user requirements'.

Personas are often named (e.g. "Lecturer Olivera" – note the inverted commas) to emphasise *human-centred* design thinking. However, they do not represent a single (named) user but cumulative characteristics of a group of people (i.e. in a certain professional role) whose decision-making needs have been considered.

Your tasks:

Step 1: Design of personas

You are required to design three personas to represent three archetypes of stakeholders (decision-makers) who would benefit from visual exploration of data from the given sources.

To create these personas, listen to the client's presentations, ask questions and research the roles, decision needs, current initiatives, future strategies, mission, vision, and aspirations. Then, design three client personas and describe them using the template shown by Figure 1.

To help you to get started with persona design, in a BUSA3021 tutorial (Week 5) we will design a practice persona "BUSA3021 Student Vijaya". This practice persona should not be included in your chosen personas.

You are required to consider 3 personas. The first two personas are given, as shown by Figures 2 and 3, and the third one is selected by your group.

For all three personas, you are required to state three non-trivial questions they will be able to answer by using your DV application. For example, a question such as "Visualise all member organisations located in Sydney" is trivial. Non-trivial questions have a potential to offer to the client something useful in the context of their work, interesting, insightful, creative and even unexpected. Non-trivial questions may require you to consider additional sources of publicly available data and combine them with your main data sets.

Persona	<"name" of your chosen Persona 1>
Role	<clearly state their role>
Context of their work/role	<describe the organisational/industry/societal context of this persona's role>
Key activities/ decision-making needs and interests	<describe their key activities, decision making-needs and interests, which are relevant for this project.
DV Questions	<three questions this particular persona would like answered through visual data exploration: Question 1: Question 2: Question 3:

Figure 1: An example of a design template

You are required to consider the following 3 personas:

Persona 1	'Australian CME Leader Melina'
Role	CEO of BCCM
Context of their work/role	The CEO of BCCM, who is engaged in value-creating opportunities for its Australian member organisations and public advocacy on behalf of Australian CMEs AND The Australian representative in the ICA who is engaged in knowledge sharing, innovation and representation of the Australian CMEs sector in the ICA.
Key activities/ decision-making needs and interests	Interested in having a visual data exploration environment of the ICA World Cooperative Monitor. The visual data exploration environment will enable finding about international ICA members and their activities that are relevant for BCCM's Australian initiatives and future strategy.
DV Questions	<three non-trivial questions about ICA Monitor this persona would like answered through visual data exploration: Question 1: Question 2: Question 3:

Figure 2: Persona 1

Persona 2	'CME Data Analyst Karl'
Role	BCCM Data analyst
Context of their work/role	- In charge of the annual BCCM NME Report - Engaged in the full life cycle of NME reporting, from data harvesting, cleansing and production of the annual NME report.
Key activities/ decision-making needs and interests	-Interested in having a visual data exploration environment of the annual NME reports, serving different stakeholders interested in the Australian CME sector.
DV Questions	<three non-trivial questions about NMEs this persona would like answered through visual data exploration: Question 1: Question 2: Question 3:

Figure 3: Persona 2

Persona 3:
This persona is chosen by your group, based on your own interests in CMEs. Make sure that this persona is relevant to the client. Personnas not related to the CMEs and not relevant to the client will not be accepted. Personnas created in the previous semester, which were related to the Co-op Federation (different client) are not acceptable either.

IMPORTANT: This persona needs to be different from the other two personas, with no overlapping or similar questions. It could be based on the same data sets as the other two.

Step 2: Analyse and Harvest Data

In this step, you need to "harvest" data from the given source of data and store the resulting data records Excel files. Instead of being given a ready-made data set in Excel, you need to figure out how to create it in the most effective way. The resulting Excel file needs to show metadata as well as the corresponding records of at least 40 organisations for the groups of 5 students or 50 for the groups of 6 students (otherwise your data will not be valid).

Note that Steps 1 and 2 are **often iterative rather than sequential**. In other words, your data collection in Step 2, may inspire new possibilities of interesting questions in Step 1, leading to additional data being collected (Step 2).

Step 3: Design of a DV application (visual data exploration environment) to answer the stated questions

Using the Excel file from Step 3 import your data set (metadata and the actual records) into Tableau, PowerBI (or other DV software platform – **please check with UC before selecting other platforms**). Tableau is highly recommended due to the functionality it provides. However, you are welcome to use other DV software platforms that offer similar functionality, so you do not need to write programming code.

IMPORTANT: You are required to **use a single platform across** all three personas. Assignments where different personas are supported by different platforms will not be accepted.

After importing your data into the DV software platform, proceed to answer all questions for the three personas, as stated in your design templates (in Step 1).
Note that Step 3 may initiate another cycle of iterative design, prompting you to go back to Steps 1 and 2 in order to expand or refine them.

Step 4: Reflection-on-action

This important step requires you to "step-back" and consider your design in broader organisational and societal context, in order to recognise and prevent possible syntactic, semantic and pragmatic data quality issues, data ethics and visual ethics issues as well as recognise and prevent potential unintended data harm. Careful consideration of these issues is of the outmost importance to the client, to our university, and to your future professional practice in any context.

This step includes the following considerations:

- **Data quality**: Discuss your approach to handling data quality, including prevention of syntactic, semantic and/or pragmatic data quality issues. Advise the client about potential DQ issues they need to be aware of.
- **Data ethics:** Use the PAPA framework (covered in class) to identify and analyse any data-related ethical issues that need to be considered in your project. Even if you think that your use of data and implementation of the DV application do not pose any ethical risks, you are still required to demonstrate your analysis of ethical issues and inform the client about your findings.

- **Visual ethics:** Discuss your approach to handling any potential visual ethics issues in your project. Advise the client about any potential visual ethics issues that could arise during their own visual data exploration (e.g. use of time filters).
- **Application of "Do No Harm" principles and guidelines in data visualisation**: Use the provided "Do No Harm" resources (posted on iLearn) to identify different guidelines relevant to your project and discuss their application. Our university is committed to equity and diversity in all aspects of our work, so your project needs to demonstrate our shared commitment by considering these recently published guidelines.

Deliverables: Project files and Group report

Group report:
The following **three files** need to be included in your group project submission (via iLearn):

- File 1: **one** zip file of the complete project, which needs to include the Excel data set and the executable DV project file in Tableau (or other DV platform – please check with UC first)
- File 2: Group report (one per group, with all **contributing** group members' names and student ids listed on the cover page)
- File 3: Group presentation – ppt slide deck (one per group) with all **contributing** group members' names and student ids listed on the cover page).

Presentation:
Up to 6 selected presentations will be presented to the client in Week 13 lecture. To be considered, your group needs to send/give an EIO to the UC by the end of Week 11. If more than 6 EIOs are received, the teaching team will make the final selection of the best 6 groups.
ALL other groups will present F2F in their allocated Week 13 tutorials.
Important: There will be NO recorded presentations. Presentations (individual or group) will not be possible at the later date, even in case of special considerations approved.

Structure of the group report.

Your group report should include the following components:

1. Introduction - a brief description of your project and its envisaged value for the client (e.g. **Why** would they use your DV application?)

2. Design templates for three personas (i.e. 3 tables) clearly showing the corresponding non-trivial questions (Step 1)

3. A sample of your resulting data set (i.e. the actual Excel records). Print screen a fraction of your Excel file showing metadata and a sample of record & paste it into your report. Full data set will be submitted in a separate Excel file.

4. Visual answers for all stated questions (Step 3) – please use print-screen to capture your resulting visualisations and provide them as Figures (one per questions).

For example:
4.1. Persona 1: <'name' of Persona 1>
4.1.1. Question 1: <state the question>
Visual answer:
<provide print-screen of your answer>
Figure x: A visual answer showing …
4..1.2. Question 2: <state the question>
Visual answer:
<provide print-screen of your answer>
Figure y: A visual answer showing …
4..1.3. Question 3: <state the question>
Visual answer:
<provide print-screen of your answer>
Figure z: A visual answer showing …

4.2. Persona 2: <insert a client persona 2>
4.2.1. Question 1:
… and so on.

5. Analysis of data quality, data and visual ethics and unintended data harm (Step 4)

6. Recommendations & Future work: Conclude your project by providing **several recommendations** to the client about possible future use and management of your visual data exploration environment beyond this project and offer some ideas for further development and innovation (e.g. additional data sets they may consider).

7. References
Please remember to reference the source of your main data set and any other public data sets used in your project. If you are using any academic/industry literature, please make sure you reference it here, using a correct referencing style.

TIPS & FAQs (OPTIONAL)

- No detailed instruction for font type, size, line space or format. As final-year University students and BA professionals-in-making, you are expected to deliver a professional piece of work that is clear, neat and well organised. Don't forget to include page numbers. Figures and tables should be always numbered and captioned, and you should refer to them at least once in the body of your writing. Using justified text is visually appealing.
- This is a PACE unit and the client may request your report and the presentation slides. Make sure that both are of the professional standard as you will not have an opportunity for post-submission/post-presentation improvements. Make sure that you do not use any copyrighted images, even those available on the client's web site.
- Clarity of language, layout and general presentation is an assessable aspect of the assignment.